Apply the Double-Angle Formulas

Formulas for $\sin 2\theta, \cos 2\theta,$ and $\tan 2\theta$ may be obtained by applying sum and difference formulas.

$$\sin 2\theta = \sin(\theta+\theta) = \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta \qquad \sin(u+v) = \sin u\cos v + \cos u\sin v$$

$$\cos 2\theta = \cos(\theta+\theta) = \cos\theta\cos\theta - \sin\theta\sin\theta = \cos^2\theta - \sin^2\theta \qquad \cos(u+v) = \cos u\cos v - \sin u\sin v$$

$$\tan 2\theta = \tan(\theta+\theta) = \frac{\tan\theta+\tan\theta}{1-\tan\theta\tan\theta} = \frac{2\tan\theta}{1-\tan^2\theta} \qquad \tan(u+v) = \frac{\tan u+\tan v}{1-\tan u\tan v}$$

$$\sin 2\theta = 2\sin\theta\cos\theta \qquad \tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

Double-Angle Formulas

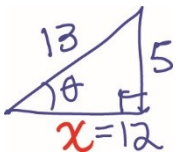
$$\cos 2\theta = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1$$

The double-angle formulas can also be used with angles other than 2θ.

For example: $\cos 4\theta = 1 - 2\sin^2 2\theta$ $\qquad \sin A = 2\sin\frac{A}{2}\cos\frac{A}{2}$

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1. Given $\sin\theta = \frac{5}{13}$ and $\tan\theta < 0$, find $\sin 2\theta$.



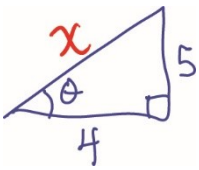
$$\left.\begin{aligned} x^2 + 5^2 &= 13^2 \\ x^2 &= 169 - 25 \\ x^2 &= 144 \\ x &= 12 \end{aligned}\right\} \quad \begin{aligned} \sin\theta &= \frac{5}{13} \\ \cos\theta &= -\frac{12}{13} \end{aligned}$$

$\sin\theta > 0, \tan\theta < 0 \rightarrow$ QII $\rightarrow \cos\theta < 0$

$$\sin 2\theta = 2\sin\theta\cos\theta = 2\cdot\frac{5}{13}\left(-\frac{12}{13}\right) = -\frac{120}{169}$$

2. Given $\tan\theta = -\frac{5}{4}$ and θ in Q IV, find $\cos 2\theta$.

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$$\left.\begin{aligned} x^2 &= 5^2 + 4^2 \\ x^2 &= 25 + 16 \\ x^2 &= 41 \\ x &= \sqrt{41} \end{aligned}\right\} \quad \cos\theta = \frac{4}{\sqrt{41}}$$

$\tan\theta < 0$, QIV $\rightarrow \cos\theta > 0$

$$\begin{aligned} \cos 2\theta &= 2\cos^2\theta - 1 \\ &= 2\left(\frac{4}{\sqrt{41}}\right)^2 - 1 \\ &= 2 \cdot \frac{16}{41} - 1 = \frac{32}{41} - 1 \\ &= -\frac{9}{41} \end{aligned}$$

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Verify the identity.

$$\frac{\cos 2x}{1-\sin 2x} = \frac{\cos x + \sin x}{\cos x - \sin x}$$

$$\text{RHS: } \frac{(\cos x + \sin x)}{(\cos x - \sin x)} \cdot \frac{(\cos x - \sin x)}{(\cos x - \sin x)}$$

$$= \frac{\cos^2 x - \sin^2 x}{\cos^2 x - 2\cos x \sin x + \sin^2 x}$$

$$= \frac{\cos 2x}{1 - 2\cos x \sin x}$$

$$= \frac{\cos 2x}{1-\sin 2x} \quad \text{(LHS)}$$

3.

$$\frac{\sin 3x}{\sin x} = 4\cos^2 x - 1$$

$$\text{LHS: } \frac{\sin(2x+x)}{\sin x} = \frac{\sin 2x \cos x + \cos 2x \sin x}{\sin x}$$

$$= \frac{2\sin x \cos x \cos x + \left(2\cos^2 x - 1\right)\sin x}{\sin x}$$

$$= \frac{2\cos^2 x \sin x + 2\cos^2 x \sin x - \sin x}{\sin x}$$

$$= \frac{4\cos^2 x \sin x - \sin x}{\sin x}$$

$$= \frac{\sin x\left(4\cos^2 x - 1\right)}{\sin x}$$

$$= 4\cos^2 x - 1 \quad \text{(RHS)}$$

4.

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Apply the Power-Reducing Formulas

The double-angle formulas for cosine can be used to express even powers of sine and cosine in terms of the first power of cosine only.

Power-Reducing Formulas: $\sin^2\theta = \dfrac{1-\cos 2\theta}{2}$ $\cos^2\theta = \dfrac{1+\cos 2\theta}{2}$ $\tan^2\theta = \dfrac{1-\cos 2\theta}{1+\cos 2\theta}$

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5. Write $\sin^4 2x$ in terms of first powers of cosine.

$$\sin^4 2x = \left(\sin^2 2x\right)^2$$

$$= \left(\frac{1-\cos 4x}{2}\right)^2 = \frac{1}{4}\left(1-2\cos 4x+\cos^2 4x\right)$$

$$= \frac{1}{4}\left(1-2\cos 4x+\frac{1+\cos 8x}{2}\right) = \frac{1}{4}-\frac{2}{4}\cos 4x+\frac{1}{8}+\frac{1}{8}\cos 8x$$

$$= \frac{3}{8}-\frac{1}{2}\cos 4x+\frac{1}{8}\cos 8x$$

Apply the Half-Angle Formulas

Half-angle formulas can be derived from the power-reducing formulas.

Substituting $\dfrac{\alpha}{2}$ for θ in the power-reducing formulas results in the half-angle formulas.

Half-Angle Formulas $\sin\dfrac{\alpha}{2} = \pm\sqrt{\dfrac{1-\cos\alpha}{2}}$ $\cos\dfrac{\alpha}{2} = \pm\sqrt{\dfrac{1+\cos\alpha}{2}}$

$$\tan\frac{\alpha}{2} = \pm\sqrt{\frac{1-\cos\alpha}{1+\cos\alpha}} = \frac{1-\cos\alpha}{\sin\alpha} = \frac{\sin\alpha}{1+\cos\alpha}$$

The sign + or − is determined by the quadrant of angle $\dfrac{\alpha}{2}$.

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6. Use the half-angle formula to find the exact value of $\sin\dfrac{5\pi}{12}$.

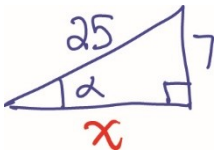
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$$\frac{5\pi}{12} = \frac{1}{2}\left(\frac{5\pi}{6}\right)$$

$$\sin\frac{5\pi}{12} = \text{In QI, choose } + \sqrt{\frac{1-\cos\frac{5\pi}{6}}{2}} = \sqrt{\frac{1+\frac{\sqrt{3}}{2}}{2}\cdot\frac{2}{2}} = \sqrt{\frac{2+\sqrt{3}}{4}} = \frac{\sqrt{2+\sqrt{3}}}{2}$$

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7. If $\sin\alpha = -\frac{7}{25}$, $\frac{3\pi}{2} < \alpha < 2\pi$, then find the exact value of $\tan\frac{\alpha}{2}$.



$$x^2 + 7^2 = 25^2$$
$$x^2 = 625 - 49$$
$$x^2 = 576$$
$$x = 24$$

QIV $\rightarrow \cos\alpha > 0$

$$\cos\alpha = \frac{24}{25}$$
$$\sin\alpha = -\frac{7}{25}$$

$$\tan\frac{\alpha}{2} = \frac{\sin\alpha}{1+\cos\alpha} = \frac{-\frac{7}{25}}{1+\frac{24}{25}} \cdot \frac{25}{25}$$
$$= \frac{-7}{25+24}$$
$$= -\frac{7}{49}$$
$$= -\frac{1}{7}$$

8. Verify the identity. $\cos^2\frac{\alpha}{2}\tan\frac{\alpha}{2} = \frac{1}{2}\sin\alpha$

$$\text{LHS: } \left(\pm\sqrt{\frac{1+\cos\alpha}{2}}\right)^2 \cdot \frac{\sin\alpha}{1+\cos\alpha} = \frac{1+\cos\alpha}{2} \cdot \frac{\sin\alpha}{1+\cos\alpha} = \frac{\sin\alpha}{2} \quad \text{(RHS)}$$

± will not matter because of the power 2

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